

Week 3 Assignment

Repository Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone>

Report Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone/blob/main/notebooks/main.ipynb>

Due to our data size not being the biggest and having to do a training, validation, and test split, we decided to do a 60/20/20 split accordingly, using a stratified random split on status, which is our target variable.

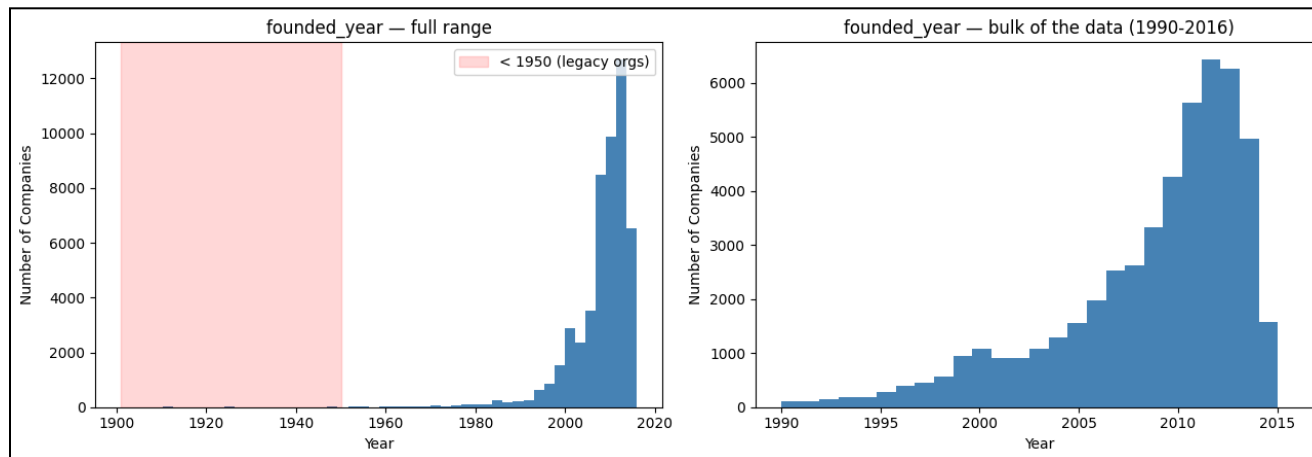
We decided to do stratification and by status because our target variable is imbalanced. The four outcome classes (operating, acquired, closed, ipo) are not evenly represented, with ipo in particular making up a small share of the data. A plain random split risks producing a validation or test set with too few examples of a minority class purely by chance, which would make performance metrics on that class unreliable and could mislead model selection. Stratifying on status ensures each of the three sets preserves the same class proportions as the full dataset, so performance differences we observe reflect the model, not sampling noise.

Our initial thoughts for partitioning were startup category and founding year, and we decided against using either as the primary split. Splitting by category would test whether a model trained on some categories generalizes to entirely unseen ones. That is not our goal. We want a model that predicts outcomes across all categories, not one intentionally blind to a subset of them. It is also structurally awkward, since category_list is multi label (a company can belong to several categories at once), so a clean partition along category lines is not possible without leaking information across the split anyway.

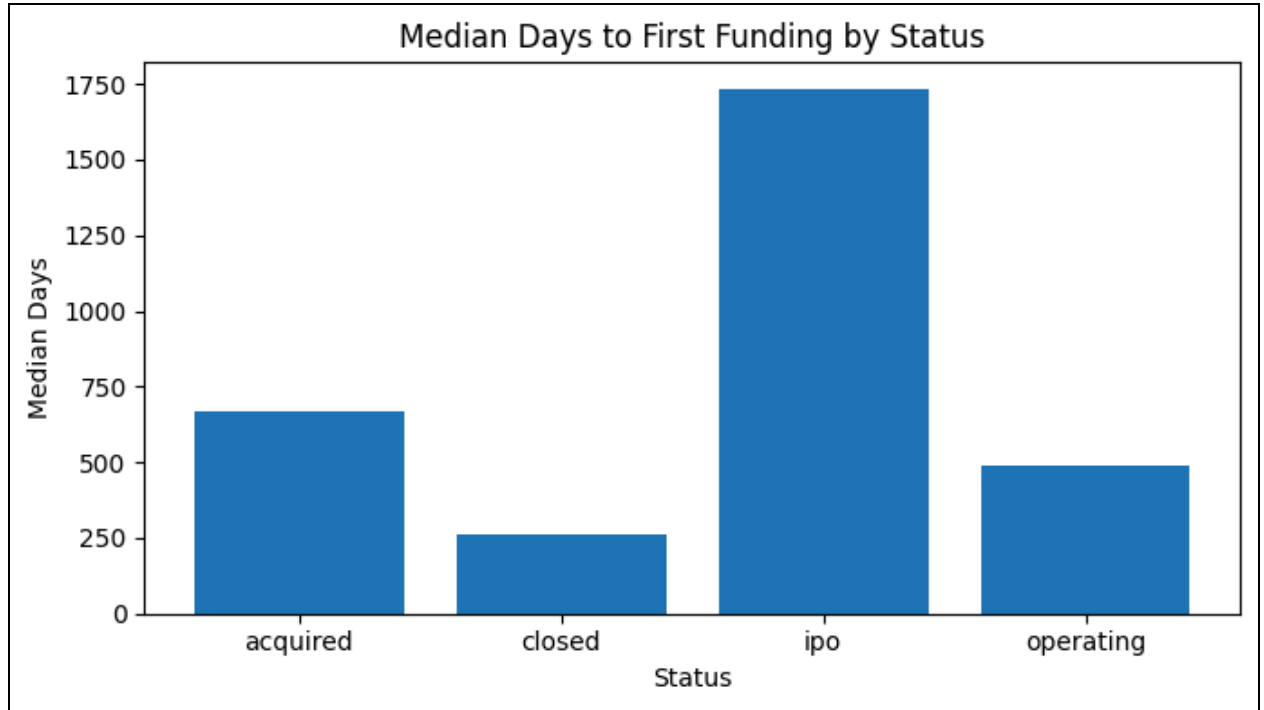
Splitting by founding year would more directly test temporal generalization, which is relevant given a subtlety in our target. Newer startups labeled operating have not necessarily succeeded. They simply have not had time to be acquired, closed, or go public yet, while older startups labeled operating carry a stronger signal. A purely random split mixes these two into every set, which could let a model lean on shortcuts such as recently founded therefore operating, without this being exposed during evaluation. We decided not to use a temporal split for our primary train, validation, and test partition, since it would reduce the

training set to older cohorts only and complicate class balance, but we are treating it as a planned follow up robustness check. This means training on earlier founding cohorts and evaluating on the most recent ones, to see whether performance degrades in a way a random split would not reveal.

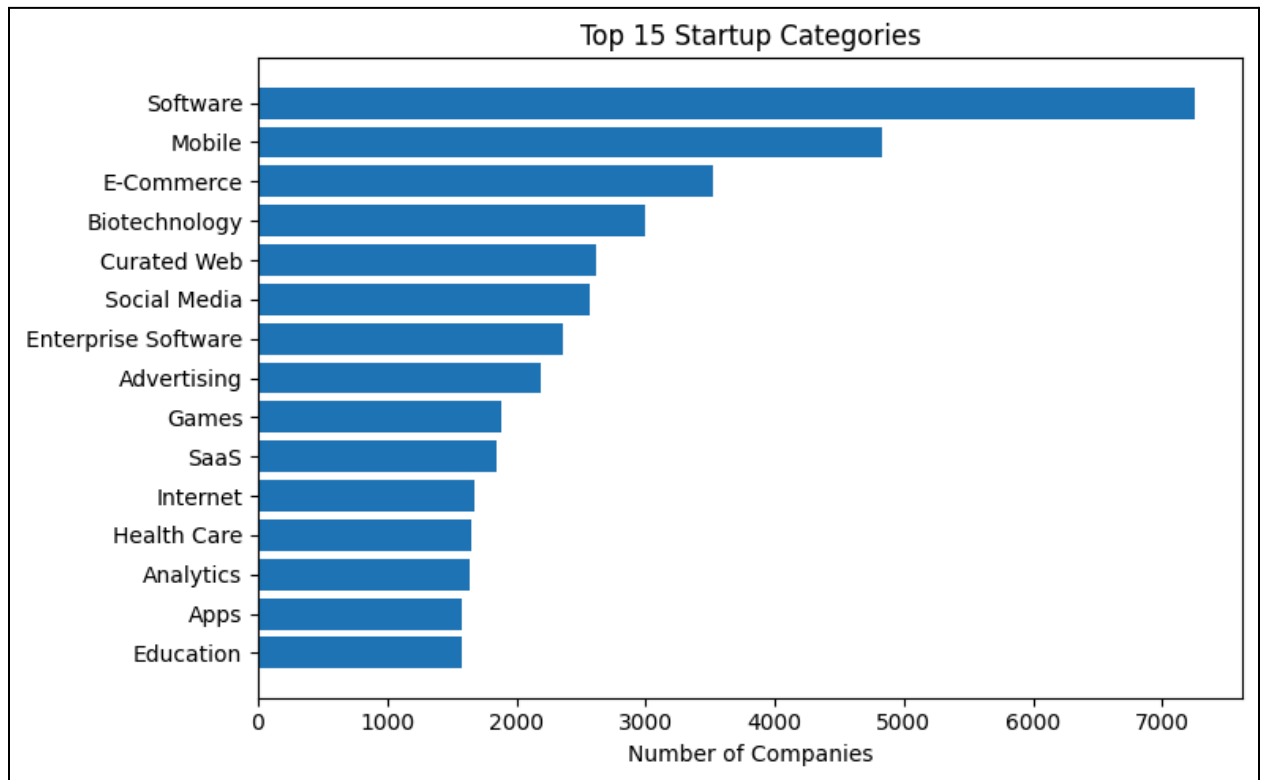
To begin the exploratory data analysis, we first had to do some data cleaning since **funding_total_usd** was stored as text and used dashes instead of blank cells for missing values. In the raw dataset, about 19% of rows fell into that bucket. Once that was numeric, we moved on to parsing the date columns (**founded_at**, **first_funding_at**, **last_funding_at**) and used them to build two new funding-velocity features, **days_to_first_funding** and **funding_span_days**, which tie back to our Week 1 idea that how fast and frequently a company raises money matters more than just how much it actually raises. While digging into the dates, we ran into another data quality issue: some rows showed funding happening before the company was even founded, and a chunk of rows turned out to be "legacy organizations" like Harvard and Purdue Research Foundation that clearly don't belong in a startup dataset. So we made the call to restrict the dataset to companies founded between 1990 and 2016, which cut about a quarter of the rows but got rid of that noise and kept things focused on the modern startup era. In terms of how quickly the startup market has been moving anyway, keeping the focus on more recent startups will be much more valuable than looking at companies from more than 75 years ago.



From there, we shifted into looking at how things break down by our target variable, **status**. We found that companies that eventually IPO'd tend to take way longer to get their first funding and raise a lot more money overall compared to companies that just stayed operating or closed down.



We also parsed out the **category_list** field (since each company can have multiple categories crammed into one string) to see which industries show up most, and Software, Mobile, and E-Commerce came out on top.



Lastly, we checked geography and duplicates in the data. The United States makes up about 59% of the dataset, and while there weren't any duplicate permalinks, we did find about 146 duplicate company names that we'll need to sort through before modeling.

The EDA showed us both opportunities and challenges for predicting startup outcomes. The biggest opportunity is that our funding-velocity features show clear differences across statuses, which supports our decision to focus on velocity. The median days to first funding increases steadily from closed (262 days) to operating (486 days) to acquired (667 days) to IPO (1,734 days), meaning closed companies got their first funding the fastest. One possible explanation is that the companies that IPO'd spent longer developing their business before seeking outside funding. Another is that they were founded earlier, so the pattern could be a result of when a company was founded rather than how quickly it moved. For example, a company that was founded in the 1990s would have decades to raise its first round, while a company founded in 2014 would only show about two years before the data ends. We will need to account for the company's age before we are able to determine if velocity is driving this difference.

The biggest challenge, however, is how imbalanced our target variable is. Around 80% of companies in the full dataset are still operating, while about 8% were acquired, 9% closed, and only 2.3% IPO'd. With this imbalance, our model could reach around 80% accuracy by predicting "operating" for every company without correctly identifying a single acquisition, closure, or IPO. So, accuracy alone doesn't tell us much here, and we will need to look at how well the model does on each outcome individually. This imbalance is also why we used a stratified split, as mentioned earlier.

Beyond class imbalance and company age, the EDA also showed some challenges with the problem statement itself. "Operating" isn't really a final outcome. A company founded in 2015 that's still operating in the data hasn't had time to get acquired, IPO, or close yet. As mentioned earlier, we plan to test for this by training on older companies and evaluating on the most recent ones. We also noticed that total funding differs a lot by status, with median funding increasing from closed (\$825K) to operating (\$1.5M) to acquired (\$10M) to IPO (\$42.8M). However, part of this gap could be a result of the outcome, since companies that survive longer have more time to raise money. This makes total funding a less reliable predictor, and also supports our decision

to focus on velocity instead. Our `days_to_first_funding` velocity feature avoids this problem, since it's measured early in a company's life, before the outcome has had a chance to affect it. However, our other velocity feature, `funding_span_days`, is more limited. About 59% of companies only had one funding round, which means their `funding_span_days` is 0, so this feature will mostly be useful for the 41% that raised more than once.

The exploratory data analysis that we conducted showed us several acute data integrity, temporal and scooping challenges that could negatively impact downstream model validity. More importantly, the dataset suffers from temporal inversions and data anomalies show in over 3,700 companies recording their funding rounds prior to their formal founding dates. Additionally, corrupted time stamps play a huge role widening the failure standard. We also identified an acute scope contamination conflict where century old institutions like Harvard and Purdue were mixed in with modern startups. While filtering out the data to companies founded between 1990 and 2016; we removed these legacy organizations, throwing out 25% of our dataset. The reasoning behind this being nearly 23% of the rows were simply missing a founding date, not because they were actually old companies. On top of that, 19% of the funding values were hidden as text dashed instead of real missing values. This skews the funding numbers by massive multi-billion dollar outliers.

We also noticed clear roadblocks in our features and the target outcome. Almost 60% of the startups raised only one round of funding, leaving their funding span at 0 days. The industry column is another challenge, since it contains 851 unique category tags all condensed together. Most importantly, the target column itself is heavily unbalanced. About 80% of the companies are labeled as operating, while only 2% reach an IPO. This introduced a big censoring problem, showing us that operating cannot be a final outcome. A newer startup is often still operating simply because it has not had enough years to either fail or exit, which could easily mislead our model.

In order to clean and prepare this data, our preprocessing will focus on saving usable rows, normalizing skewed numbers and simplifying the complex features. Instead of eliminating the companies with missing founding dates, we can estimate their founding year based on their first round of funding. We can adjust small negative gaps to zero to account for money raised before formal incorporation. The money columns can be converted from dashes to proper null values, with a true or false flag added. This will track undisclosed funding. Lastly, we should turn the single round issue into a simple yes or no feature, and reduce the 851 categories down to the top 25 most common industries. We can apply

balanced weight classes when training the model so the model pays fair attention to rare outcomes like an acquisition or an IPO.